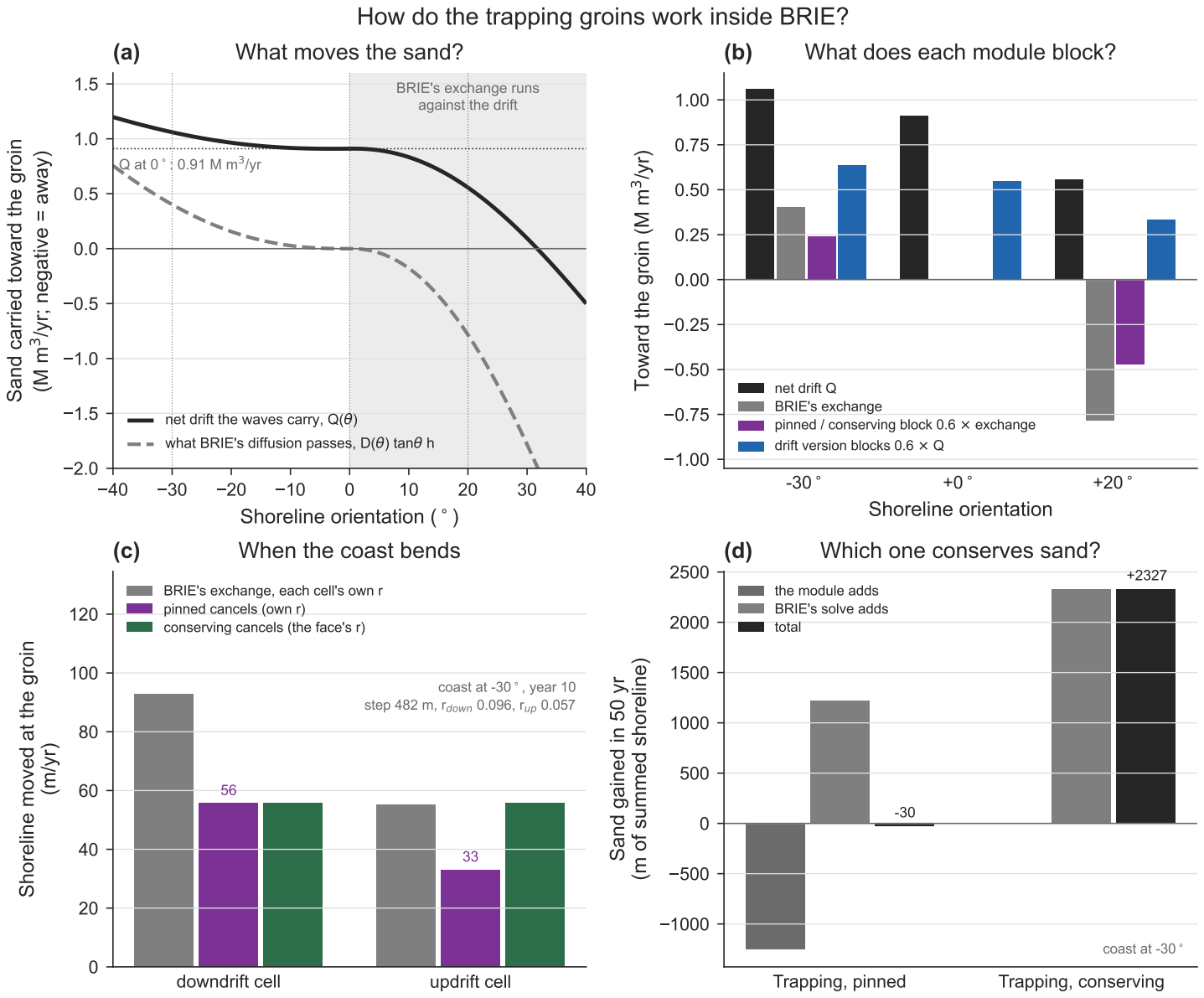
How do the trapping groins work inside BRIE?
(a)
What moves the sand?
BRIE's exchange runs against the drift
Q at 0 ° : 0.91 M m³/yr
net drift the waves carry, Q(θ)
what BRIE's diffusion passes, D(θ) tanθ h
Sand carried toward the groin (M m³/yr; negative = away)
Shoreline orientation (°)
(b)
What does each module block?
net drift Q
BRIE's exchange
pinned / conserving block 0.6 × exchange
drift version blocks 0.6 × Q
Toward the groin (M m³/yr)
-30 °
+0 °
+20 °
Shoreline orientation
(c)
When the coast bends
BRIE's exchange, each cell's own r
pinned cancels (own r)
conserving cancels (the face's r)
coast at -30 °, year 10
step 482 m, r_down 0.096, r_up 0.057
56
33
Shoreline moved at the groin (m/yr)
downdrift cell
updrift cell
(d)
Which one conserves sand?
the module adds
BRIE's solve adds
total
+2327
-30
Sand gained in 50 yr (m of summed shoreline)
coast at -30 °
Trapping, pinned
Trapping, conserving